

Lecture 11 Tutorial — AI Ethics and Future Development

Q1. AI Lifecycle and Iteration (Ordering + Selection)

The lecture describes an AI system workflow with iterative refinement. Arrange the stages in the correct order:

- A. Collect Data
- B. Feature Extraction
- C. Machine Learning
- D. System Deployed
- E. Formulate Problem

Write the correct order as a sequence (e.g., $E \rightarrow A \rightarrow \dots$).

If the deployed system “does not work”, which **TWO** actions are explicitly suggested in the lecture? (Choose two)

- (i) Fine tune
- (ii) Refine feature
- (iii) New data
- (iv) Refine the problem
- (v) Reduce sample size

Q2. Concerns vs Practices (Matching)

Match each problem to the most directly corresponding practice listed in the lecture.

Problems:

1. Data privacy
2. Data collection bias (background/population mismatch)
3. False media
4. Power consumption of the system

Practices:

- A. Deepfake detection
- B. Anonymisation (really work?) / federated learning idea
- C. Know your population (particularly background information)
- D. Algorithm simplification

Write: 1-_____, 2-_____, 3-_____, 4-_____.

Q3. Federated Learning and Anonymisation (Multiple Choice)

Which statement best matches the lecture's motivation for "learning with anonymisation (federated learning)"?

- A. Centralising data always reduces leakage risk.
- B. Transmitting data introduces a risk of data leaking; keeping data local reduces central exposure but may introduce inconsistency issues.
- C. Federated learning guarantees identical feature definitions and distributions across clients.
- D. Federated learning removes all forms of bias by design.

Q4. Data Collection Bias Mechanisms (Select all that apply)

When data are stored on local machines (as discussed in the lecture), which issues may occur? Select all that apply.

- A. Inconsistent feature definition (or data collection)
- B. Inconsistent data quality
- C. Inconsistent feature distribution
- D. Perfect matching of training and ground-truth distributions
- E. Automatic elimination of domain shifts

Q5. Dataset vs Ground Truth Distribution (True / False + One-line Fix)

Write T or F. If false, rewrite the statement correctly in one line.

1. The training dataset can exactly match the ground-truth probability distribution if the sample size is large enough.
2. If we cannot cover all varieties, different circumstances can be viewed as different domains, motivating domain generalisation/adaptation.

Q6. Domain Shift Responses (Matching)

Match each term to the best description according to the lecture context.

Terms:

- A. Domain generalisation
- B. Domain adaptation
- C. Transfer learning

Descriptions:

1. Learn from a source domain and adjust to a target domain with distribution shift.
2. Aim to perform well on unseen domains without direct target-domain tuning.
3. Reuse knowledge/representations from one task/domain to help learning in another.

Write: A-_____, B-_____, C-_____.

Q7. Algorithm Bias as Assumption Mismatch (Matching)

The lecture states that using an algorithm introduces assumptions. Match each algorithm to its stated assumption.

Algorithms:

- A. GMM
- B. SVM
- C. Naive Bayes

Assumptions:

- 1. Impacts of factors are independent of each other
- 2. There exists a feature space that may classify data with a linear boundary
- 3. All clusters can be approximated by Gaussian distributions

Write: A—_____, B—_____, C—_____.

Q8. Evaluation Bias and Metric Choice (Multiple Choice)

The lecture reminds you about confusion-matrix-related metrics. Choose the best metric for each requirement.

- 1. “Overall performance across all classes”
 - A. Accuracy
 - B. Precision
 - C. Recall
- 2. “Quality of positive predictions (minimise false positives)”
 - A. Accuracy
 - B. Precision
 - C. Recall
- 3. “Ability to identify the positive class (minimise false negatives)”
 - A. Accuracy
 - B. Precision
 - C. Recall

Q9. Distance Measures (Matching)

Match each distance to what it measures as stated in the lecture.

Distances:

- A. Euclidean distance
- B. Cosine distance
- C. Correlation distance

Measures:

- 1. Absolute distance
- 2. Directional distance
- 3. Temporal distance

Write: A—_____, B—_____, C—_____.

Q10. Security Threats to AI Systems (Classification)

The lecture lists actions that can mislead an AI system.

Classify each scenario as **Perturbation** or **Data poisoning**.

1. Adding small, visually imperceptible noise to an image at test time to flip the prediction: _____
2. Injecting a small number of maliciously crafted training samples into the dataset to degrade performance: _____
3. Adding barely audible noise to an audio clip at inference time to cause mis-transcription: _____
4. Planting mislabeled samples into the training set so a trigger pattern causes a targeted misclassification: _____

Q11. Energy / Sustainability Calculation (Computation)

The lecture provides the comparison:

Human (Jie Ke): 500 KCal $\approx$ 0.0005 kWh

AlphaGo: 60 kWh

1. Compute the ratio

$$\frac{60}{0.0005}.$$

2. Interpret the result: AlphaGo consumed approximately _____ times the energy (in kWh units) of the human reference.